On April 20, 2016, Opera became the first major browser maker to integrate an unlimited and free VPN (Virtual Private Network). Now, if you use Opera browser, you don't have to download the VPN extension/add-on or pay for any VPN subscription to secure your online browsing activity on public Wi-Fi.

With a free, unlimited, native VPN that doesn't require any subscription fee, Opera wants to make VPNs available to everyone, and provide privacy and security.

What are Opera extensions?

Opera extensions are code packages that you can add to your Opera Web browser to extend its functionality. The Opera extension API provides functions to explore different features of the Web browser by giving you a variety of options for the extension functionality. If you have already developed extensions for Google Chrome, then you will encounter some similar steps and procedures while developing extensions for Opera. Since Opera is based on Chromium, a majority of core internal functionalities are similar to the Google Chrome Web browser, which means it is possible to run some of the Chrome extensions on Opera.

Now, let us create a privacy based search extension for the Opera Web browser.

Why do we require privacy on search?

When we perform a search on search engines and click on our search results, our search terms or key words are sent to the website we clicked on (via an HTTP referrer header). This concept of sharing personal information is called 'search leakage'. In simple words, when we search for something private, we are sharing that private search information with our search engine as well as with the websites that we clicked on for that search. Also, when we visit any website after that, our computer automatically sends important information about the websites we visited along with our IP address and user agent, to that clicked website. So, basically, when we perform a search, other websites know our search terms and they also know that we searched for them.

Some search engines save our history, along with the date and time of the search, as well as some important computer information such as our user agent, IP address or our account information (name and login information, if we are logged in).

This availability of information about us does raise some privacy concerns. Thankfully, there are some search engines that avoid these privacy and security problems. The three most popular ones are listed here.

DuckDuckGo: This is one of the most popular Internet search engines, with its main motto being to protect searchers' privacy and to avoid the creation of personalised search results for users. DuckDuckGo puts privacy first and does not log user information such as IP addresses, etc. It also provides full protection from search leakage.

Encrypted Google: This uses SSL (Secured Socket Layer), which encrypts the connection between your computer and Google. This basically prevents and protects it from others, like Internet service providers and public Wi-Fi hotspots, from seeing your search results page and other vital information.

Ixquick: This lets you search privately using your favourite search engines. It gathers results from all major search engines. It adds an extra security layer between you and your search engines, which means search engines will see search requests coming from Ixquick and not the end user. This makes the user's search requests practically anonymous to the main search engine. Ixquick does not log search terms, IP addresses or any other personal information about users.

So now, let us develop an extension, which will allow a user to search any highlighted text or word, on the DuckDuckGo, Encrypted Google and Ixquick search engines.

Developing an extension

We are now going to develop a user-friendly and useful privacy based search extension. Let us call it the 'Privacy Search' extension, which will allow the user to search any highlighted word on one of these privacy based search engines—duckduckgo.com, encrypted.google.com and ixquick.com. Users will highlight the word they wish to search in the Web browser, then will right-click the highlighted word and select the 'Privacy Search' option from the context menu. Following this, three sub-menu options will be displayed: 'Search on DuckDuckGo for..', 'Search on Encrypted Google for..' and 'Search on Ixquick for..' Users will select and click on any one of these sub-menu options, depending on which search engine they want to use. After doing this, users will be redirected to the highlighted word's search results page, opened in a new tab.

We are going to use the following Web technologies:

- JSON to create a manifest file
- JavaScript for the logic

Setting up the development environment

Let us set up a file structure for our extension, and create a new folder with any name on the computer. In our case, *C:\privacysearch* is the extension's folder. Now, inside that new folder, create a folder called 'images'. In the root directory of our extension folder, create an empty text file using Notepad and save it with the *.json* extension called *manifest.json* and also create an empty JavaScript file called *background.js*.

You should end up with the following directory structure:

- images\
- manifest.json
- background.js